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LAU
School of Engineering

ELE 537, Communication Systems
Exam 3, Fall 2013
Instructor: Dr. Chadi Abou-Rjeily

Problem 1: (10 Points; (3,3), (4,1))

The Signal-to-Noise Ratio (SNR) of a Delta-Modulated (DM) system with audio signals is given by:

$$\left(\frac{S}{N}\right) = \frac{3f_s^3 APPR}{(1600\pi)^2 B}$$

where:

- f_s is the sampling frequency.
- B is the bandwidth of the audio signal assumed to be equal to 4 kHz in what follows.
- $APPR$ is the average-to-peak power ratio assumed to be equal to 1/2 in what follows.

a. Find the value of the sampling frequency required to achieve a SNR of 40 dB with DM.

b. Deduce the bit rate of the DM system.

Consider a PCM system transmitting at the same rate as the above DM system.

c. Find the SNR of the PCM system if the analog signal is sampled at the Nyquist rate.

d. Determine whether DM or PCM is better in this case.

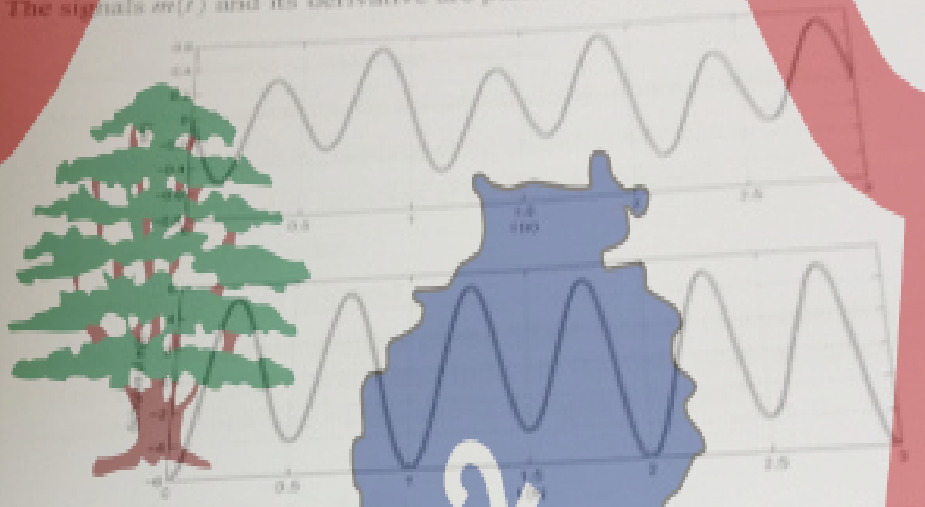
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Problem 2: (7 points)
Delta-Modulation (DM) is applied on the signal given by:

$$m(t) = -\frac{1}{2\pi}\sin(2\pi t) - \frac{5}{4\pi}\sin(4\pi t)$$

The signals $m(t)$ and its derivative are plotted in the figures below:



If $m(t)$ is sampled at 5 times the Nyquist rate, what is the step size δ required to prevent the slope-overload noise while minimizing the granular noise?

Problem 3: (5 points)
Consider a filter whose equivalent transfer function is shown in the figure below:



Determine wether the above filter satisfies the Nyquist first method for eliminating the Inter-Symbol-Interference (ISI) at a baud rate of 15 kbauds (*attention*: the x-axis in the above figure is labeled in kHz).

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Problem 4: (12 points)
The purpose of this problem is to design TDM systems that allow to accommodate the following three sources:

- One analog source of bandwidth 2 kHz.
- One analog source of bandwidth 6 kHz.
- One digital source of data rate 8 kbits/s.

You are allowed to use only the following components in your design:

- One commutator that revolves at 4,000 revolutions/s.
- One commutator that revolves at 8,000 revolutions/s.
- One 2-bit PCM encoder.
- Pulse stuffing is not allowed.

Provide the two possible designs of the TDM system (for each design, draw a block diagram for your system and indicate the data rates at various points on your diagram). Which one of the above designs is better?

Problem 5: (10 points: (2,4,4))
An FM radio is tuned to receive an FM broadcasting station at 90 MHz. The radio is of the superheterodyne type with the Local Oscillator (LO) operating on the high side of the 90-MHz input and using a 10.7 MHz Intermediate-Frequency (IF) amplifier.

a. Determine the frequency of the LO.

b. If the FM signal has a bandwidth of 200 kHz, give the requirements of the RF and IF filters.

A second FM radio station is broadcasting at 111 MHz and the corresponding transmitted FM signal has a bandwidth of 200 kHz.

c. Will the signals transmitted by the above radio stations interfere with each other at the output of the superheterodyne receiver?

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Problem 6: (18 points: (2, 2, 3, 10))
The complex envelope of a modulated signal is given by:

$$g(t) = 2 + \cos(2\pi f_o t)$$

where the carrier frequency satisfies the relation: $f_c \gg f_o$.

a. Find the quadrature modulation components $x(t)$ and $y(t)$ for this modulated signal.

b. Find the magnitude and phase modulation components $R(t)$ and $\theta(t)$ for this modulated signal.

c. Write down the expression of the modulated signal, denoted by $s(t)$.

d. Find the spectrum $S(f)$ of the modulated signal by two different methods. Provide a labeled sketch of the magnitude spectrum.

Problem 7: (7 points: (3, 4))
A Phase-Modulated (PM) signal is given by:

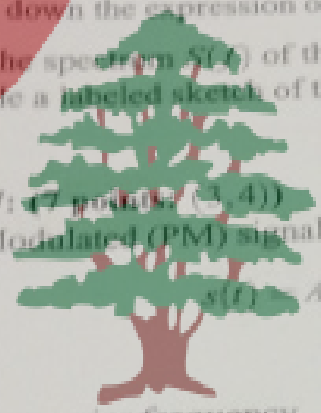
$$s(t) = A_c \cos\left(2\pi f_c t + D_p m(t)\right)$$

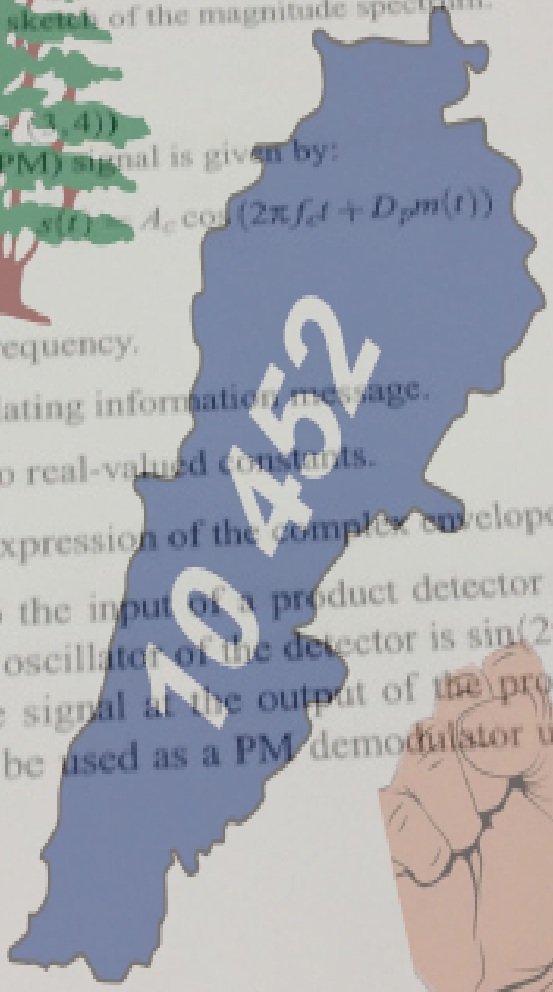
where:

- f_c is the carrier frequency.
- $m(t)$ is the modulating information message.
- A_c and D_p are two real-valued constants.

a. Write down the expression of the complex envelope $g(t)$.

b. $s(t)$ is applied to the input of a product detector where the signal delivered by the local oscillator of the detector is $\sin(2\pi f_c t)$. By evaluating the expression of the signal at the output of the product detector, show that this detector can be used as a PM demodulator under the assumption that $D_p \ll 1$.







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Problem 8: (7 points)
Consider the following bandpass signal:

$$v_{in}(t) = 10\sin(2\pi 10^3 t)\cos\left(2\pi 10^8 t + 10\cos(2\pi 10^3 t)\right)$$

This signal is applied to the input of the FM detector shown in the figure below:



Assuming that the followed envelope detector is ideal, write down the expression of the signal at the output of the above FM detector.

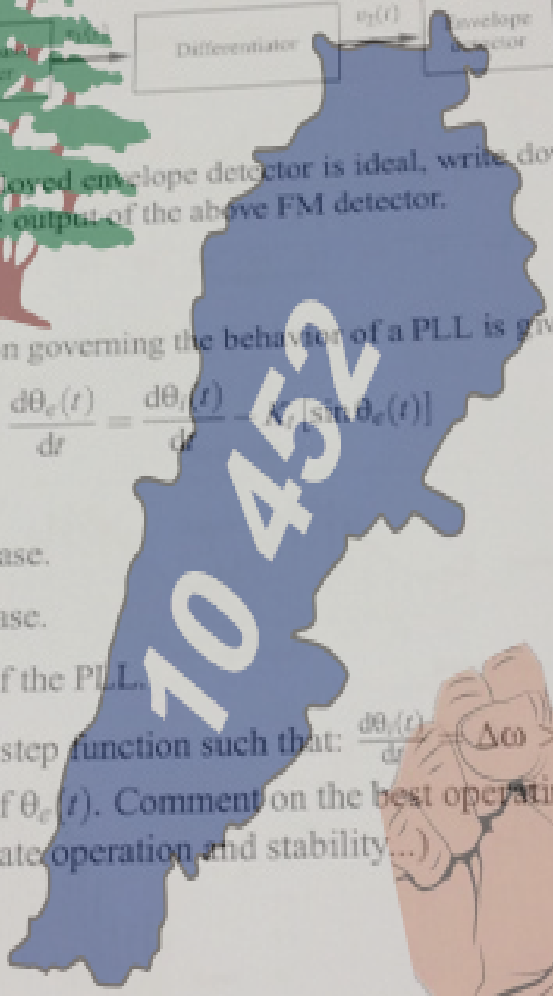
Problem 9: (7 points)
The differential equation governing the behavior of a PLL is given by:

$$\frac{d\theta_e(t)}{dt} = \frac{d\theta_i(t)}{dt} - K_t \sin[\theta_e(t)]$$

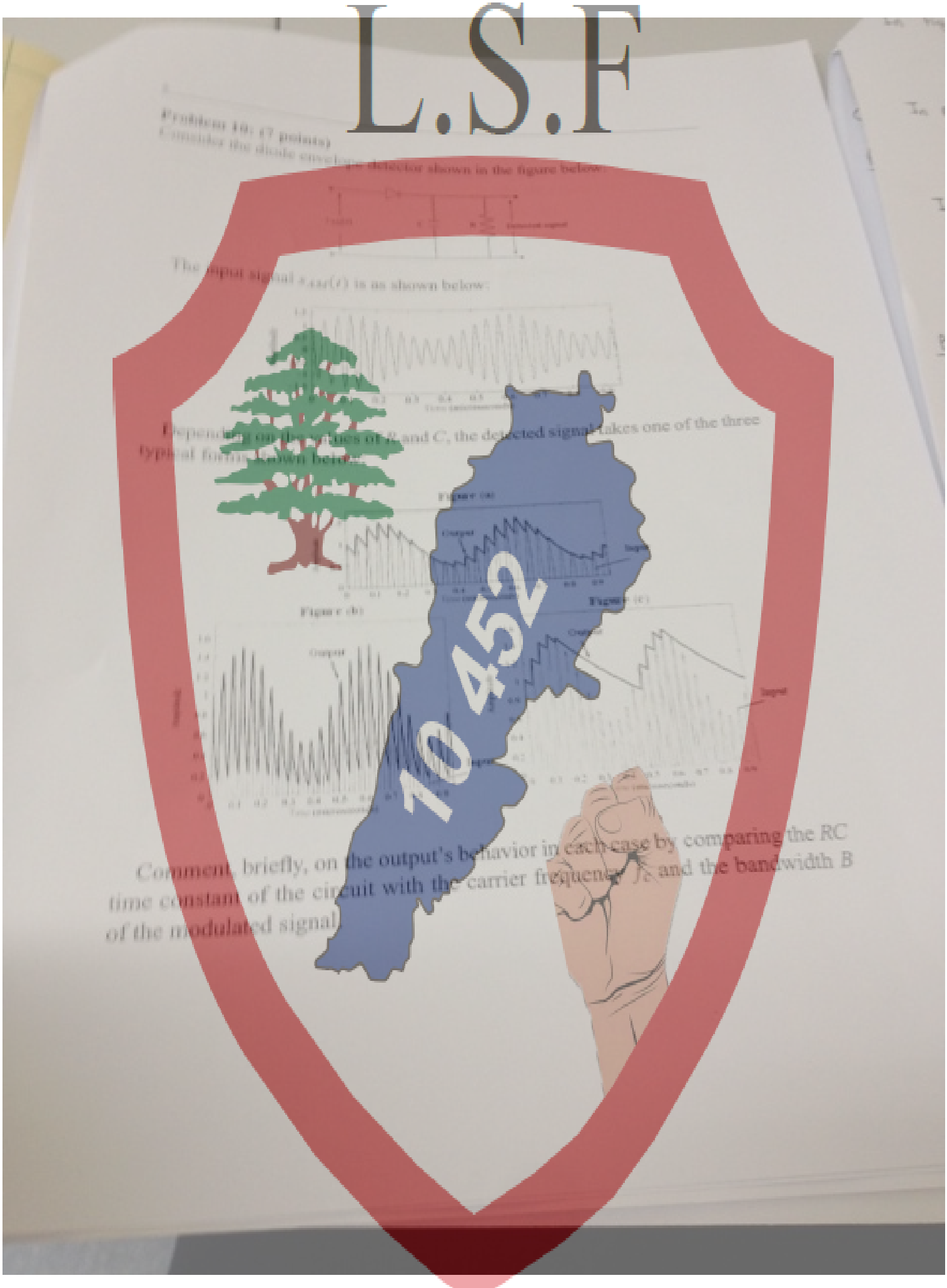
where:

- $\theta_e(t)$ is the error phase.
- $\theta_i(t)$ is the input phase.
- K_t is the total gain of the PLL.

Let the information be a step function such that: $\frac{d\theta_i(t)}{dt} = \Delta\omega > 0$ for $t \geq 0$.
Plot $\frac{d\theta_e(t)}{dt}$ as a function of $\theta_e(t)$. Comment on the best operating point (value of the error phase, steady-state operation and stability...)






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